

HACKCANTON S2 · REAL-WORLD ASSET (RWA) & BUSINESS WORKFLOWS

LedgerFactor

Confidential invoice financing on Canton. A supplier discounts a buyer-approved invoice to a financier, and the financier's margin is hidden from the buyer **by the protocol**. The ledger **structurally** guarantees one invoice can never be financed twice.

LIVE AND DAML-SCRIPT PROVEN

DAML 2.10.4 ON CANTON

A supplier waits 60 days. A financier will pay today, for a price.

Invoice factoring is a **4 trillion dollar** market that runs on one commercial secret: the discount rate the financier charges. Put that trade on a shared ledger and two things break.

BREAK 1 · PRICE LEAK

The buyer sees what the financier earns. That is **negotiating leverage against the supplier** on every future invoice, and it is the reason confidential factoring stays off-chain.

BREAK 2 · DOUBLE PLEDGE

The same receivable financed twice to two lenders. Classic factoring fraud. Off-chain it is caught by **registries and trust**, after the fact.

THE BIND

Encrypt it and you lose the shared truth that prevents the double pledge. Publish it and you lose the price. **You need both at once.**

Three guarantees, enforced by the ledger and not by the UI

SELECTIVE DISCLOSURE

The rate lives on **FinancingProposal / FinancingOffer**, whose only stakeholders are the financier and the supplier. The buyer is never a stakeholder, so the margin **never reaches the buyer's participant node**. Not hidden in a view. Not there.

SINGLE EXERCISE

The Invoice is one authoritative contract. **AcceptFinancing is a consuming choice**, so a second financing attempt on the same invoice is rejected by the ledger itself. Anti-double-pledge is structural, not a check we remembered to write.

ATOMIC DVP

Assigning the receivable and paying the supplier the discounted cash **settle in a single transaction**. No half-financed state exists.

On a public chain the sealed bid leaks through logs before the reveal, and the margin is a matter of public record. **Canton sub-transaction privacy is the product**, not a feature of it.

The 1.97% margin reaches two of four parties

SUPPLIER

Northwind Supply

Issues & discounts the receivable

Suptf3ch111g...2c39

RECEIVABLE · INVOICE

Q3 pallet delivery

\$100,000

CONFIRMEDnot yet listed

FINANCING OFFER · CONFIDENTIAL

MARGIN / DISCOUNT

1.97%

ADVANCE TO SUPPLIERFINANCIER SPREAD

\$98,030\$1,970

disclosed: financier + supplier

5 contracts on this participant's node

BUYER

Globex Corp

confirms the payable

Buylf3ch111g...2c39

RECEIVABLE · INVOICE

Q3 pallet delivery

\$100,000

CONFIRMEDnot yet listed

FINANCING TERMS

Withheld by the Canton ledger from this participant. A 1.97% margin exists on a contract this party is not a stakeholder of.

1 contract on this participant's node

FINANCIER

Meridian Capital

underwrites & funds

Finlf3ch111g...2c39

RECEIVABLE · INVOICE

Q3 pallet delivery

\$100,000

CONFIRMEDnot yet listed

AI UNDERWRITING · Q3 PALLET DELIVERY

89/100A

APPROVE

recommended discount1.97%

RELIABILITYCONCENTRATIONDILUTIONSIZE

FINANCING OFFER · CONFIDENTIAL

MARGIN / DISCOUNT

1.97%

ADVANCE TO SUPPLIERFINANCIER SPREAD

\$98,030\$1,970

disclosed: financier + supplier

2 contracts on this participant's node

AUDITOR

Regulator

observes the audit trail

Audlf3ch111g...2c39

FINANCING TERMS

Withheld by the Canton ledger from this participant. A 1.97% margin exists on a contract this party is not a stakeholder of.

0 contracts on this participant's node

One invoice. Four participant nodes,
each a **live query as that party**.

Supplier and financier see the rate.
Buyer and auditor get a **redaction bar**, because their nodes were never sent the contract.

Drive it yourself on the live site, or verify it against the running gateway with one command from the README.

Sealed-bid auction: no financier can see a rival's bid

A supplier auctions one invoice to three financiers. Each submits a **blind bid**.

As a financier, **only your own envelope opens**. Rivals stay wax-sealed. As the supplier and auctioneer, every envelope opens. Buyer and auditor see no pricing at all.

On close the **lowest bid wins** and settles atomically. Proven on-ledger by **testSealedBidAuction**.

The screenshot displays the Veild interface for a sealed-bid auction. At the top, the 'INVOICE AMOUNT' is 100000 and the 'DESCRIPTION' is 'Q3 pallet delivery'. Below this, a row of buttons shows 'OPEN', 'SEAL BIDS', 'VIEW AS...', and 'REVEAL', with 'VIEW AS...' being the active state. A 'Reset' button is on the right. The main section is titled 'VIEW LEDGER AS' and shows 'YOU HOLD ONE KEY, AS Meridian Capital FINANCIER'. Below this, a row of buttons shows 'AUCTIONEER Supplier', 'FINANCIER Meridian Capital' (active), 'FINANCIER Apex Credit', 'FINANCIER Cobalt Partners', 'OBSERVER Buyer', and 'OBSERVER Auditor'. The 'MERIDIAN CAPITAL - AGGRESSIVE BIDDER' section shows 'You see only your own envelope. Rivals stay wax-sealed - the ledger never sent you their rate.' Below this, the 'SEALED LOT - BUYER-APPROVED RECEIVABLE' section shows 'Q3 pallet delivery' and 'FACE VALUE \$100,000'. The bottom section shows three panels for the financiers: 01 Meridian Capital (active) showing a discount rate of 1.97%, advance to supplier of \$98,030, and financier spread of \$1,970; 02 Apex Credit showing a sealed bid; and 03 Cobalt Partners showing a sealed bid. A footer note states: 'Only the Supplier's master key can close this auction. Switch to Supplier to accept the lowest bid.'

Impossible on a public chain: sealed bids would leak via logs before the reveal.

An underwriting engine, not a chatbot

The agent underwrites the receivable over **only the data the financier is entitled to see**.

That is the point: the AI inherits the ledger's disclosure boundary, it does not reach around it.

DETERMINISTIC CORE

Scores buyer payment reliability, portfolio concentration and dilution risk into a **0-100 credit score**, a risk band and a recommended discount rate. **5/5 unit tests**.

HUMAN IN THE LOOP

The agent recommends. **The financier reviews and signs** the resulting on-ledger offer. The rate the demo executes is the rate the agent actually recommended.

NO NETWORK DEPENDENCY

An LLM turns the structured rationale into a short memo. With no API key it **falls back to a deterministic memo**, so the demo never depends on a network call.

Daml Script tests, each querying the ledger as its own party

```
$ daml test  
daml/Tests.daml:setupParties: ok, 0 active contracts, 0 transactions.  
daml/Tests.daml:testSelectiveDisclosure: ok, 2 active contracts, 4 transactions.  
daml/Tests.daml:testNoDoubleFinancing: ok, 5 active contracts, 11 transactions.  
daml/Tests.daml:testHappyPathSettlement: ok, 3 active contracts, 7 transactions.  
daml/Tests.daml:testSealedBidAuction: ok, 6 active contracts, 12 transactions.
```

DISCLOSURE

The buyer queries for FinancingOffer and gets **nothing**. The financier and supplier see the margin.

DOUBLE FINANCE

Two offers on one invoice. The first AcceptFinancing consumes it, the second is **rejected by the ledger**.

SETTLEMENT

Atomic DvP: supplier paid **97,000 on a 100,000 invoice**, original receivable gone, auditor sees face value only.

What is real, and what is not

The privacy and the atomicity are real and proven. Two things are deliberately scoped, and we would rather say so than be found out.

MOCK CASH

Cash is an owner-signed bearer token standing in for a token-standard holding. The DvP **atomicity is real**; the cash model is not production grade. Real token-standard DvP under **CIP-0056** is the stretch goal.

NOTIFIED FACTORING

Listing an invoice reveals the financier's identity to the buyer. The sensitive figure, **the pricing, stays hidden**. Undisclosed factoring would use Canton explicit disclosure.

Everything else on these slides is running at ledgerfactor.unitynodes.com right now and reproducible from the public repo.

Built, deployed, and provable in one command

SHIPPED

Daml model with both money-shots proven by Daml Script · AI credit-scoring agent, unit-tested · live JSON API gateway with party-scoped views · four-role frontend · **Veild sealed-bid auction** · persistent deployment behind Caddy and Cloudflare

NEXT

Real token-standard DvP settlement under CIP-0056 · runtime compliance citation in the auditor view

LIVE DEMO

SOURCE, APACHE-2.0

ledgerfactor.unitynodes.com

github.com/UnityNodes/ledgerfactor